

FAHAD FARID

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SUMMARY

Data Scientist with **3+ years** across Financial Services, Insurance and Defence R&D. Builds and validates **ML and Deep Learning Models** end to end, covering Feature Engineering, Validation, **Performance Monitoring** and **Drift Root Cause Analysis**, and ships them as **Production Scoring Pipelines on GCP and BigQuery**. Designs **Controlled Experiments** to isolate **Incremental Lift**, turns model output into budget decisions and mentors juniors.

SKILLS & TOOLS

Programming & Engineering: Python (Pandas, NumPy, Scikit-learn), SQL (Large-Scale Extraction), PySpark, PostgreSQL, SQLAlchemy/Alembic, Unit Testing, Git and Peer Code Review, Airflow Orchestration, Reproducible Data Pipeline Design, Data Cleaning and Wrangling, Advanced Excel

Machine Learning: Deep Learning (TensorFlow), Scikit-learn, Feature Engineering, Model Selection and Validation, Performance Monitoring, Model Interpretability, Drift and Root Cause Analysis, NLP, Large Language Models, LLM API Integration, Prompt Engineering, HuggingFace, OpenCV

Cloud, AI & Visualization: GCP, BigQuery, Claude AI, Claude API, Claude Code, Production Scoring Pipelines, UAT and PIV Validation, Automated Scoring Repositories, Data Quality Monitoring, Streamlit, Google Maps API, QGIS, Tableau, Power BI, Executive-Level Storytelling

Statistics & Experimentation: Controlled A/B Simulations, Test vs. Control, Incrementality Measurement, Hypothesis Testing, Segmentation

PROFESSIONAL EXPERIENCE

American Express — Gurugram, India Feb 2025 – Present
Analyst, Data Science — Global Decision Science (Credit & Fraud Risk)

- Designed, built and deployed the end-to-end **Spend Model for US Consumers (USCS)**, delivering **\$1.68M in gross contribution margin** via **controlled A/B simulations** across targeted and non-targeted populations, isolating true incremental lift and reshaping budget allocation.
- Led end-to-end migration of **all 3** marketing models to **GCP and BigQuery**, rewriting code to new guidelines, applying variable logic changes, onboarding new variables and re-scoring in Airflow orchestrated, unit tested Python, UAT and PIV validated, runtime **54% faster at 13 to 6 hrs**.
- Simulated **Incremental Card Member Value** from the measured lift, used by the Campaign Teams to unlock **product specific budget allocations**, and designed a **Centralized Scoring Repository** moving performance tracking from **semi-annual to monthly**, saving **30 days** every **6 months**.
- Established model and data quality monitoring across **80+ variables** and **5 marketing models** to surface drift early, and performed **Performance and Model Interpretability Tracking biannually** for **3 models** as per Governance Standards, with all performing satisfactorily.
- Performed **Root Cause Analysis** on a variable shift that breached guideline thresholds, identified the **segment level nuance** behind it and presented the justification at the CFR level **Model Development and Performance Committee (MDPC) Forum**.
- Formally **mentored and onboarded junior analysts** onto the model stack, presenting results and limitations to marketing and risk partners.

ICICI Lombard General Insurance Company — Headquarters, Mumbai Jul 2024 – Feb 2025
Senior Manager, Strategy and Innovation (AI/ML)

- Developed an **AI-powered Q&A bot** using LLMs over internal policy documentation, improving query efficiency for **400+ underwriters**.
- Identified **6 high-potential pincodes** in the UK for business expansion from **92**, combining internal portfolio data with external risk reports.
- Built QGIS Landslide risk dashboards tiering **2,000+ policies** into **5 levels**, and mapped congestion across **13 MH pincodes** from **8,700+ clients**.

Aeronautical Development Agency, DRDO, Bangalore — Junior Research Fellow Jul 2023 – Jun 2024
Research Experience at Ministry of Defence cum M.Tech Thesis — Prof. N. S. Vyas, IIT Kanpur

- Led **Digital Twin Development** for IVHM of Combat Aircraft using flight data from **180+ sorties** across **20+ aircraft**, improving efficiency.
- Built DL models for **7 systems** including FCS ($R^2 = 0.97$), BMS ($R^2 = 0.93$) and PS ($R^2 = 0.91$), with above **95% fault validation** on held out data.
- Enabled **real-time condition-based maintenance**, presented at **MadeAI 2024, Porto** and won **Best Paper (Innovation)** at **AeroCON, India**.
- Published as "Deep Learning-Based Digital Twinning Models for Inter System Behavior and Health Assessment of Combat Aircraft Systems", **SAE Technical Paper 2024-26-0478**, AeroCON 2024 — doi.org/10.4271/2024-26-0478

EDUCATION

Indian Institute of Technology Kanpur (IIT Kanpur) — M.Tech, Solid Mechanics and Design | CPI 8.25/10 Jul 2022 – Aug 2024
Jamia Millia Islamia, New Delhi — B.Tech, Mechanical Engineering | CPI 9.06/10 Aug 2017 – Jul 2021

PERSONAL PROJECTS

Job Search and Matching Agent | Python, PostgreSQL, SQLAlchemy, Claude API, Apify, Streamlit Aug 2026

- Built an end-to-end ingestion pipeline screening around **200 LinkedIn postings a day** against deterministic role and eligibility rules, then grading each against a candidate profile with a **Claude API match score** and surfacing the **top 20** for application, automating a previously manual search.
- Listed all **200 daily postings** on a **Streamlit dashboard**, backed by a **normalised PostgreSQL schema** with SQLAlchemy and Alembic migrations, data freshness states, company alias resolution, API retry handling and human review gates before any application is surfaced for review.

AWARDS & RECOGNITION

- Top Contributor Award**, American Express (Mar 2026), for the GCP migration of **all 3 marketing models**.
- Extra Mile Award**, American Express (Sep 2025), for GLIDE utility testing and score based budget allocation.
- Track Best Paper Award (Innovation)**, SAE AeroCON International Conference (Jun 2024).